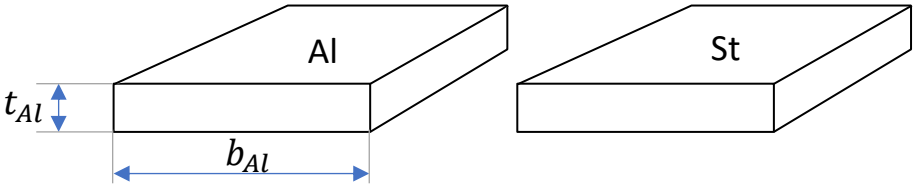
Structures and Materials

Tutorial | Dr.-Ing. Andreas Hermanutz

Structural Design Elements

1. Aluminium vs. Steel Plate



Task 1: Compare the mass and bending stiffness of an aluminum plate and a steel plate that have been designed to possess identical tensile stiffness.

A) Calculate the thickness ratio between the plates necessary to ensure their tensile stiffnesses are equal.

$$EA \qquad E_{St}A_{St} = E_{Al}A_{Al} \qquad E_{St}t_{St}b = E_{Al}t_{Al}b \qquad \frac{\boldsymbol{t_{Al}}}{\mathbf{t_{St}}} = \frac{E_{St}}{\mathrm{E}_{Al}} = \frac{210\ GPa}{70\ GPa} = \frac{3}{1}$$

B) Calculate the mass ratio of the two plates, given that they have identical tensile stiffness.

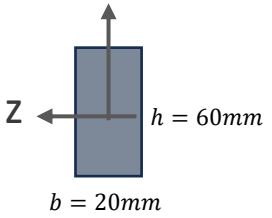
$$\frac{\boldsymbol{m_{Al}}}{\mathbf{m_{St}}} = \frac{\rho_{Al} \cdot t_{Al} \cdot L \cdot b}{\rho_{St} \cdot t_{St} \cdot L \cdot b} = \frac{2.7\frac{kg}{m^3} \cdot 3}{7.8\frac{kg}{m^3} \cdot 1} = 1.04$$

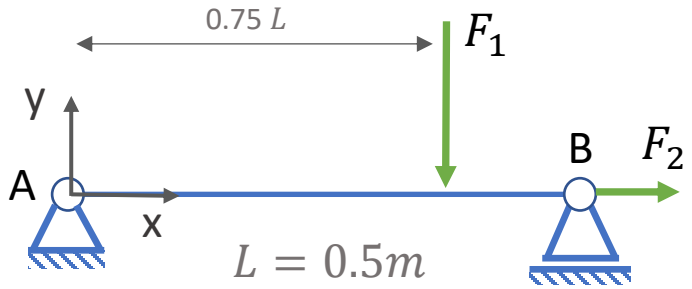
C) Calculate the bending stiffness ratio of the two plates, given that they have identical tensile stiffness.

$$\boldsymbol{EI} \qquad I_{rectang.} = \frac{bt^3}{12} \qquad \frac{\boldsymbol{E_{Al}I_{Al}}}{\boldsymbol{E_{St}I_{St}}} = \frac{E_{Al}}{\mathrm{E}_{St}} \cdot \frac{bt_{Al}^3}{12} \cdot \frac{12}{bt_{St}^3} = \frac{E_{Al}}{\mathrm{E}_{St}} \cdot \left[\frac{t_{Al}}{t_{St}}\right]^3 = \frac{1}{3} \cdot 27 = 9$$

2. Beam Structural Analysis

Task 2: Given is an elastic aluminium beam ($E = 70GPa$, $\nu = 0.3$) loaded by a constant force $F_1 = 10kN$ and $F_2 = 30kN$.





A) Calculate the maximum normal stress in the structure and its safety margin, if the maximum allowable stress is $\sigma_{b,allow.} = 350MPa$.

step 1: reaction forces

$$\sum F_Y: \quad A_Y + B_Y - F_1 = 0$$
$$\sum M: \quad -A_Y \cdot L + F_1 \cdot 0.25L = 0$$

$$A_Y \cdot L = F_1 \cdot 0.25L$$
$$A_Y = F_1 \cdot 0.25$$

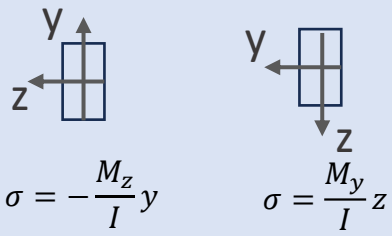
$$B_Y = F_1 - A_Y = 0.75\ F_1$$

$$A_Y = 2.5\ kN$$
$$B_Y = 7.5\ kN$$

step 2: max bending moment

$$-A_Y \cdot 0.75L + M = 0$$
$$M_{max} = A_Y \cdot 0.75L$$
$$M_{max} = 0.9375\ kNm$$

sign convention: positive bending moments induces tensile normal stress in the bottom:



$$\sigma = -\frac{M_z}{I} y \qquad \sigma = \frac{M_y}{I} z$$

step 3: bending stress

$$\sigma_{b,max} = \frac{-M_{max}}{I_z} z_{max} \qquad I_z = \frac{b \cdot h^3}{12} = 360000\ mm^4$$

$$\sigma_{b,max}\ (30) = \frac{-937.5 \cdot 10^3\ Nmm}{360000\ mm^4} 30mm = -78.125\ MPa$$

$$\sigma_{b,max}(-30) = \frac{-937.5 \cdot 10^3\ Nmm}{360000\ mm^4} (-30mm) = 78.125\ MPa$$

step 4: normal stress

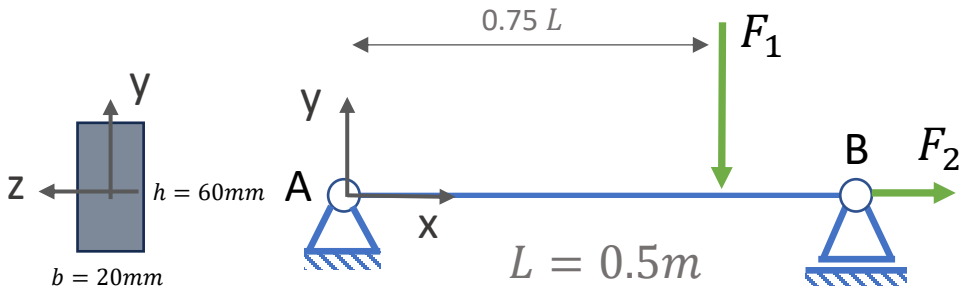
$$\sigma_N = \frac{F_2}{b \cdot h} = 25\ MPa$$

step 5: $\sigma_N + \sigma_{b,max} = 103.125\ MPa$

$$SF = \frac{\sigma_{b,allow.}}{\sigma_{max}} = \frac{350}{103.125} = 3.4$$

2. Beam Structural Analysis

Task 2: Given is an elastic aluminium beam ($E = 70GPa$, $\nu = 0.3$) loaded by a constant force $F_1 = 10kN$ and $F_2 = 30kN$.



B) Include the transfers shear force and check, if the beam is sufficiently dimensioned considering the von Mises criterion $\sigma_V = 350MPa$.

$$\boldsymbol{\sigma_V = \sqrt{\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2}}$$

$$\boldsymbol{\sigma_y = \sigma_z = 0\ MPa}$$

$$\boldsymbol{\sigma_V = \sqrt{\sigma_x^2 + 3\tau_{xy}^2}}$$

$$\tau_{max} = \frac{Q_z \cdot h^2}{I_z \cdot 8} = \frac{A_Y \cdot h^2}{I_z \cdot 8} = 9.375MPa$$

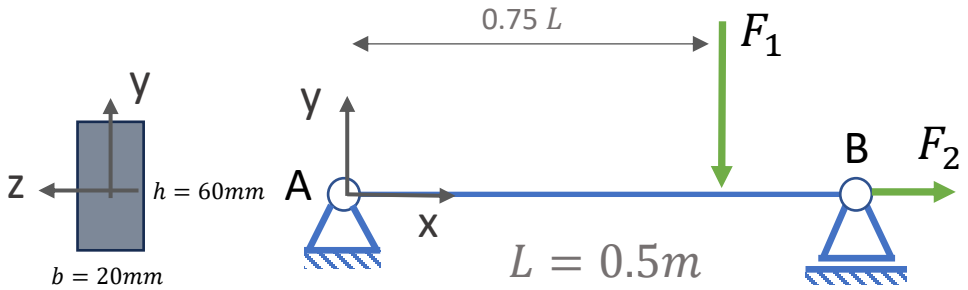
$$\sigma_V = \sqrt{(103.125)^2 + 3(9.375)^2} = 104.4\text{MPa}$$

$$SF = \frac{\sigma_{b,allow.}}{\sigma_{max}} = \frac{350}{104.4}$$

no relevant change in the safety factor

2. Beam Structural Analysis

Task 2: Given is an elastic aluminium beam ($E = 70GPa, \nu = 0.3$) loaded by a constant force $F_1 = 10kN$ and $F_2 = 30kN$.



C) Calculate the deflection of the beam considering shear deformation.

$$w''(x) = \frac{-M(x)}{EI}$$

section 1

$$M(x) = A_Y\, x$$

$$w'(x) = \frac{-A_Y}{2EI}x^2 + C_1$$

$$w(x) = \frac{-A_Y}{6EI}x^3 + C_1 x + C_2$$

$w(x=0) = 0 \quad \rightarrow C_2 = 0$

$$w'\left(x = \frac{3l}{4}\right) = 0 \quad \rightarrow C_1 = \frac{9\,A_Y\,l^2}{16\,EI}$$

$$w\left(x = \frac{3l}{4}\right) = \frac{-A_Y}{6EI}\left(\frac{3l}{4}\right)^3 + \frac{9\,A_Y\,l^2}{16\,EI}\left(\frac{3l}{4}\right)$$

$$= \frac{-A_Y}{6EI}\left(\frac{3l}{4}\right)^3 + \frac{27\,A_Y\,l^3}{64\,EI}$$

$$= \frac{-A_Y l^3}{EI}\left(\frac{27}{6\cdot 64} + \frac{27}{64}\right)$$

$$w_b = \frac{-A_Y l^3}{EI}\frac{63}{128} = -6.1\ mm$$

$$w_s' = \frac{Q}{\kappa GA} = \frac{-A_Y}{\kappa GA}$$

$$G = \frac{E}{2(1+\nu)} = 26923MPa$$

$$\kappa \approx 5/6$$

$$w_s = \frac{-A_Y}{\kappa GA}x + C_1 \qquad w(x=0) = 0 \rightarrow C_1 = 0$$

$$w_s = w_s\left(x = \frac{3l}{4}\right) = -0.035mm$$

$$w_{total} = w_b + w_s = -6.135\ mm$$